

3. Application Under Test (AUT)

3.1 Introduction

Application Under Test (AUT) refers to the software application on which testing activities are performed. In this project, the BlazeDemo Flight Booking Application was selected as the AUT to evaluate its performance under simulated user load conditions.

The application provides a complete flight reservation workflow that allows users to search for available flights, select a preferred flight, enter passenger details, and complete the booking process. This workflow makes BlazeDemo suitable for demonstrating performance testing concepts and evaluating application behavior under concurrent user activity.

The primary objective of testing this application is to measure response times, throughput, success rates, and overall system stability while multiple virtual users interact with the system simultaneously.

3.2 Application Name

Application Name: BlazeDemo Flight Booking Application

BlazeDemo is a sample web-based airline reservation application designed for software testing and automation practice. The application simulates real-world flight booking operations and provides a simple user interface for searching and purchasing flights.

The application contains multiple interconnected web pages that represent a complete booking journey from flight search to booking confirmation.

Key functionalities include:

- Flight Search
- Flight Selection
- Passenger Information Entry
- Flight Booking Confirmation
- Reservation Processing

These functionalities were used as the basis for creating load testing scenarios in Gatling.

3.3 Application URL

Official Website:

<https://blazedemo.com>

The application is publicly accessible and does not require user registration or authentication to perform flight booking operations.

The website serves as a demonstration platform for testing various software quality assurance activities, including:

- Functional Testing
- Automation Testing
- API Testing
- Performance Testing
- Load Testing

Because of its simplicity and accessibility, BlazeDemo is widely used by software testers and automation engineers for practice and learning purposes.

3.4 Business Workflow

The BlazeDemo application follows a straightforward flight reservation process that simulates a real-world airline booking system.

The business workflow begins when a user visits the application's home page and selects a departure city and destination city. The system then displays a list of available flights matching the selected route.

The user can select one of the available flights and proceed to the booking page. On the booking page, passenger details and payment information are entered. Once the information is submitted, the system processes the booking and displays a confirmation page indicating that the purchase has been completed successfully.

The workflow tested during this project is illustrated below:

Complete Booking Workflow

Home Page



Select Departure City

↓

Select Destination City

↓

Find Flights

↓

Display Available Flights

↓

Choose Flight

↓

Enter Passenger Details

↓

Purchase Flight

↓

Booking Confirmation

3.5 Load Testing Scenarios Derived from Business Workflow

To simulate realistic user behavior, the business workflow was divided into three separate load testing scenarios.

Scenario 1 – Search Flights

This scenario focuses on the flight search functionality.

Workflow:

Home Page

↓

Select Departure City

↓

Select Destination City



Find Flights

Purpose:

To evaluate how efficiently the application retrieves and displays available flight information under load.

Scenario 2 – Choose Flight

This scenario extends the flight search process by selecting a flight.

Workflow:

Home Page



Find Flights



Choose Flight

Purpose:

To measure the application's performance during flight selection transactions.

Scenario 3 – Complete Booking

This scenario represents the complete end-to-end business transaction.

Workflow:

Home Page



Find Flights



Choose Flight



Enter Passenger Details



Purchase Flight



Confirmation Page

Purpose:

To evaluate application performance during the entire booking process and verify successful transaction completion under load conditions.

3.6 Scope of Testing

The scope of this performance testing activity includes:

- Flight Search Operations
- Flight Selection Operations
- Booking Submission Operations
- Confirmation Page Processing
- Response Time Measurement
- Throughput Analysis
- Success Rate Validation

The scope excludes:

- Database Performance Testing
- Security Testing
- Browser Compatibility Testing
- Stress Testing Beyond Defined User Load

- Infrastructure-Level Monitoring

The testing activity is focused exclusively on evaluating the application's behavior under the configured load conditions using Gatling.

3.7 Summary

The BlazeDemo Flight Booking Application was selected as the Application Under Test because it provides a complete and realistic booking workflow suitable for load testing. The application's flight search, flight selection, and booking functionalities were tested using Gatling to measure performance metrics and evaluate system stability under simulated user load conditions.